Cloud Computing & Big Data Infrastructure

Lec-0

Outline

- **Introduction**: Cloud Computing
- History and evolution
- Overview of Types of Computing
  - Cluster Computing
  - Grid Computing
  - Utility Computing
  - Autonomic Computing
- Applications of cloud computing for various industries
  - Health
  - E-Governance
  - Education
  - Agriculture
  - Geospatial Data
- Economics and benefits of cloud computing.

About Me

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About Subject

Subject Name	Cloud Computing & Big Data Infrastructure
Subject Code	CS 504
Number of Lecture Per Week	3
Number of Labs	1
Total Credits	4

Syllabus

Unit -1	**8 Hours**
Introduction to Cloud computing, Cloud computing: IaaS, PaaS, SaaS; Cloud computing: Current & future trends, Impact on scientific research Types of Cloud services providers: Google, Amazon, Microsoft Azure, IBM, Sales force	
Unit - 2	**8 Hours**
Virtualization for Cloud: Need for Virtualization, Pros and cons of virtualization, Types of virtualization; system VM, process VM, virtual machine monitor; Virtual machine properties, Interpretation and Binary translation, HLL VM – Hypervisors – Xen, VMWare, Hyper - V	
Unit - 3	**8 Hours**
Introduction to Big data. Big data and its importance, Drivers, Big data analytics, Big data applications. Algorithms, Matrix-Vector, Multiplication by Map Reduce. YARN, SQOOP, PIG	

Syllabus (Cont'd)

Unit - 4	**8 Hours**
OpenStack – Cloud operating system, object storage (SWIFT). Introduction, distributed file system. Hadoop – Architecture, Hadoop distributed file system (HDFS) MapReduce – Programming model, Task scheduling algorithms, Data serialization	
Unit - 5	**10 Hours**
Apache Spark Java AWS SDK, S3 API, Relational Database Service, SimpleDB Service, NoSQL Databses, HIVE, MongoDB, HBase Elastic search, Qlik Sense, Superset. Building Interactive Visualization using Superset.	
Total Contact Time: 42 Hours	

What is Cloud

Buyya

- Cloud is a parallel and distributed computing system consisting of a collection of inter-connected and virtualized computers that are dynamically provisioned and presented as one or more unified computing resources based on service-level agreements (SLA)

Vanquero

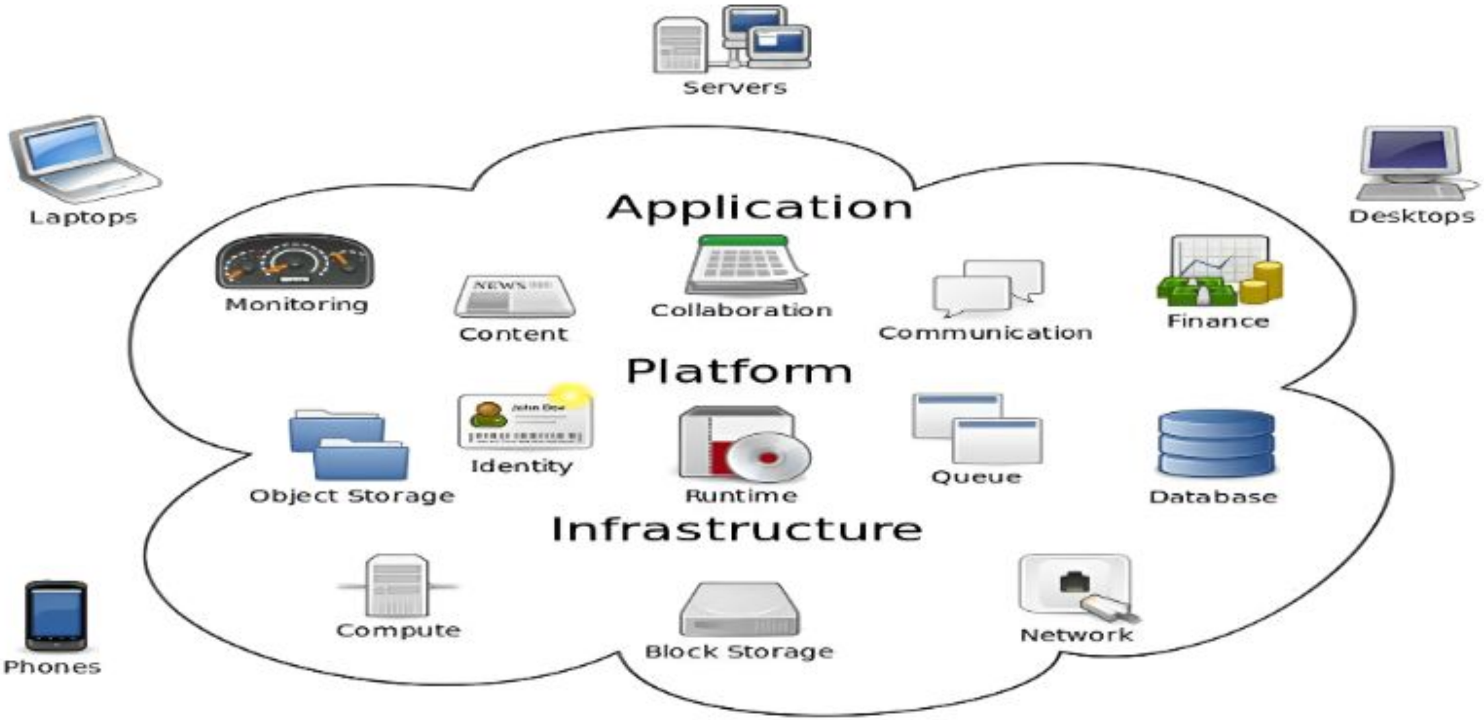
- Clouds are a large pool of easily usable and accessible virtualized resources (such as hardware, development platforms and/or services). These resources can be dynamically reconfigured to adjust to a variable load (scale), allowing also for an optimum resource utilization.

McKinsey

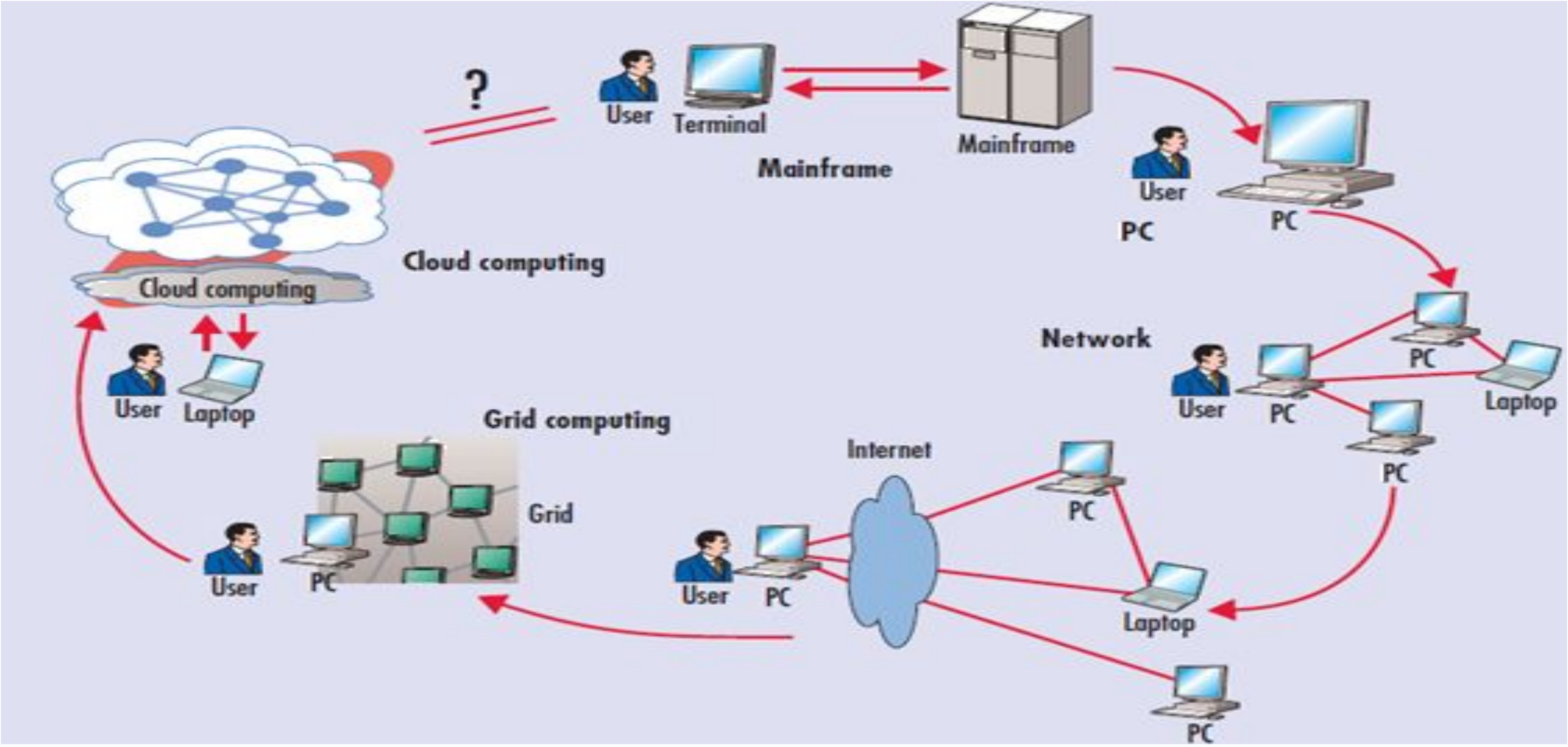
- Clouds are hardware based services offering compute, network, and storage capacity where: Hardware management is highly abstracted from the buyer, buyers incur infrastructure costs, and infrastructure capacity is highly elastic

Cloud Computing

The phrase **Cloud Computing** means a type of Internet-based computing - including servers, storage and applications.



History and Evolution



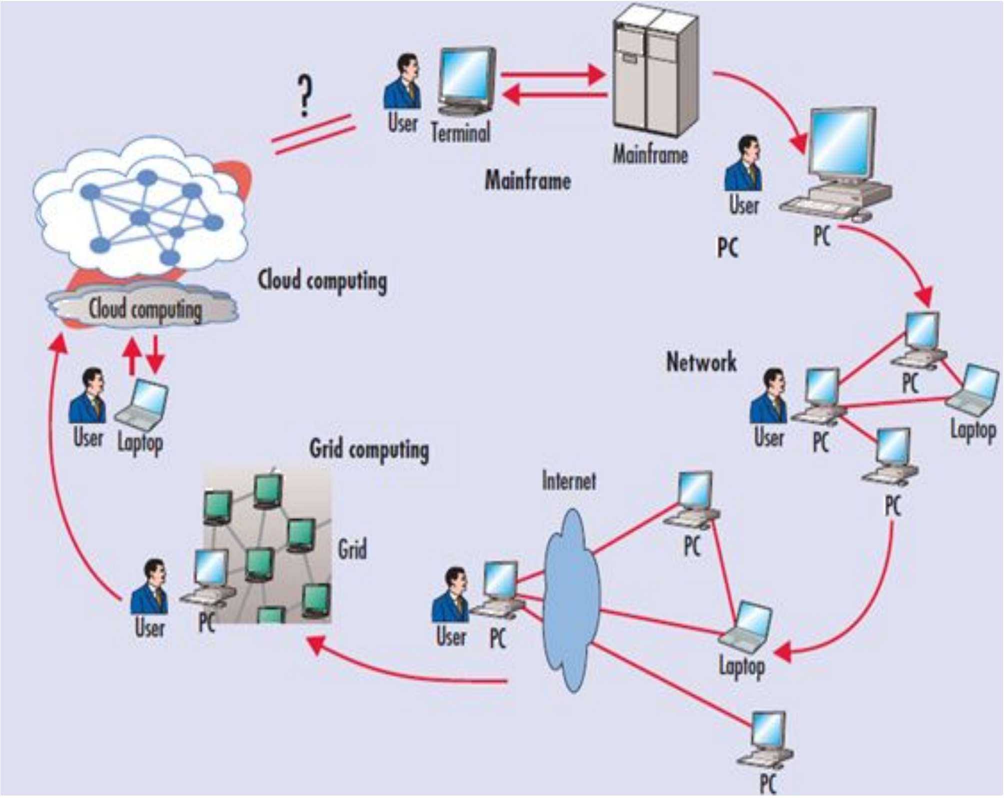
Computing paradigm shift

1969, Leonard Kleinrock

Evolution of Cloud Computing

Mainframes

- These were the first examples of large computational facilities leveraging multiple processing units.
- Powerful, highly reliable computers specialized for large data movement and massive input/output (I/O) operations.
- Not Distributed, offered large computational power by using multiple processors, which were presented as a single entity to users.

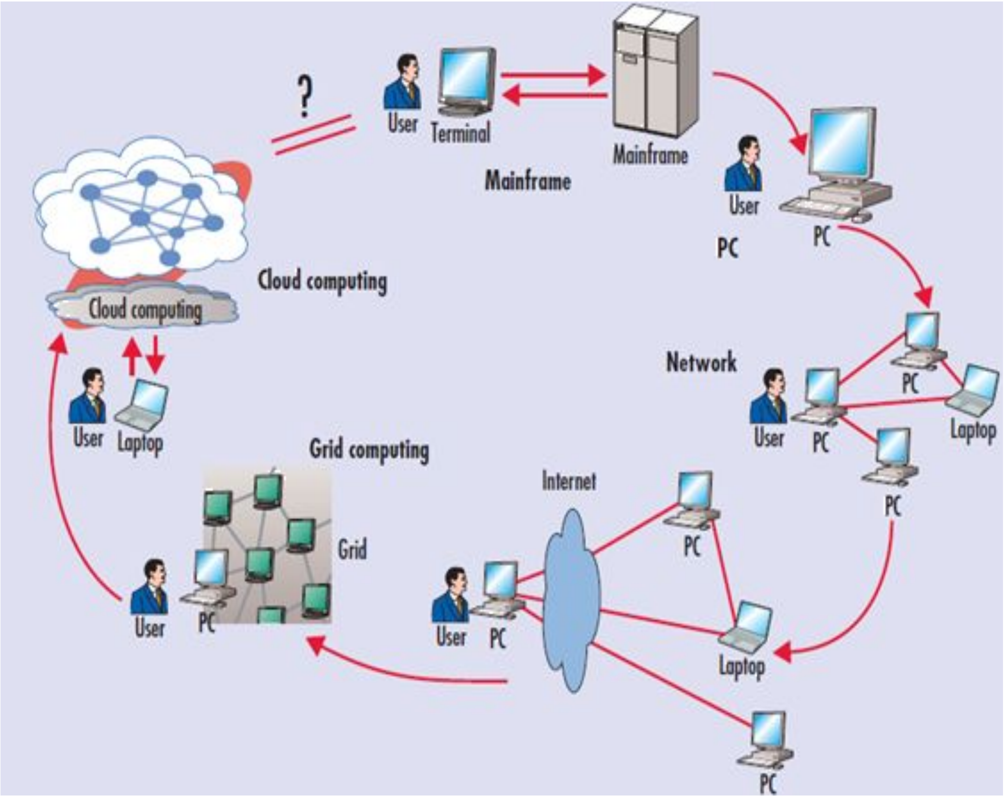


Computing paradigm shift

Evolution of Cloud Computing

Cluster Computing

- Starting in the 1980s, clusters become the standard technology for parallel and high-performance computing.
- Technology advancements created cheap commodity machines.
- Machines could then be connected by a high-bandwidth network and controlled by specific software tools that manage them as a single system.

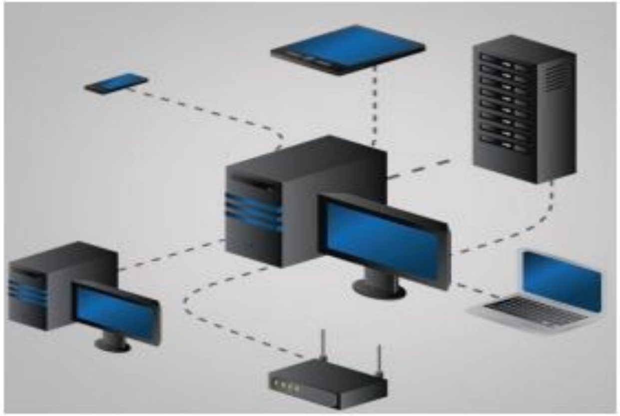


Computing paradigm shift

Cluster Computing

A cluster is a type of parallel and distributed system

It consists of a collection of inter-connected stand-alone computers.



The Cluster

A Node

A single or multiprocessor system with memory, I/O facilities, & OS

A Cluster

Generally two or more computers (nodes) connected together
In a single cabinet, or physically separated & connected via a LAN
Appear as a single system to users and applications
Provide a cost-effective way to gain features and benefits

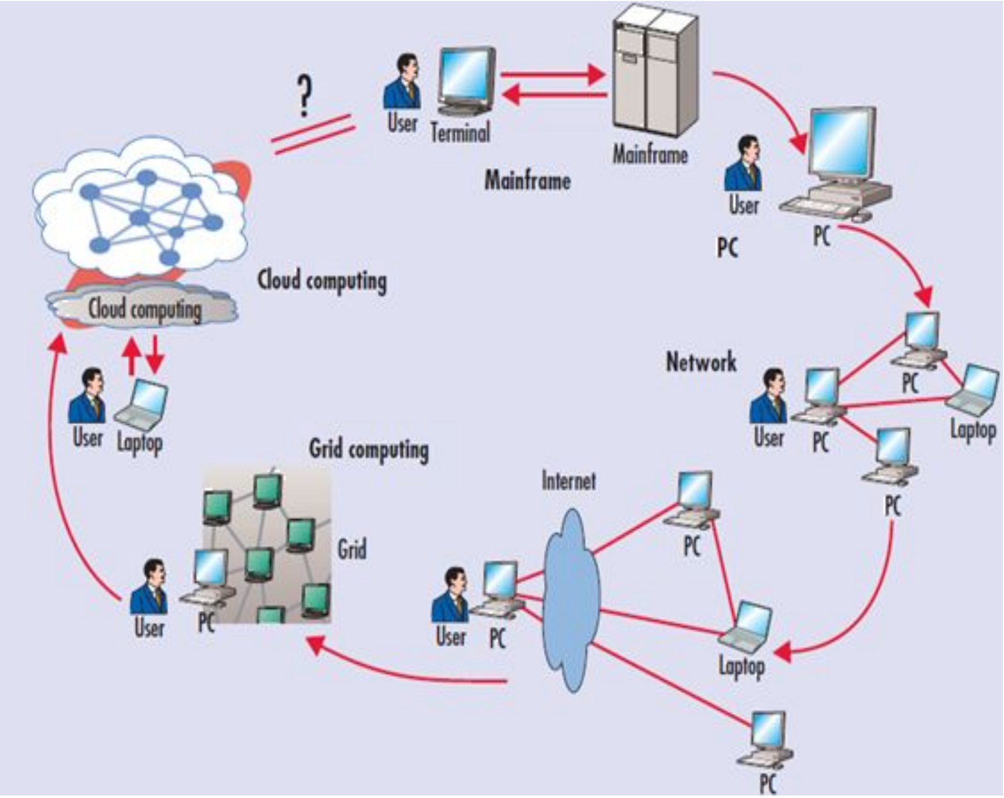
Benefits of Cluster Computing

- **System Availability**: High due to redundancy of hardware, operating system, and applications.
- **Hardware fault tolerance**: Redundancy
- **Scalability**: Adding servers to the cluster or by networking multiple clusters as the need arises.
- **High Performance**

Evolution of Cloud Computing

Grid Computing

- Appeared in the early 1990s as an evolution of cluster computing.
- Grids initially developed as Aggregations of geographically dispersed clusters by means of Internet connections.
- These clusters belonged to different organizations, and arrangements were made among them to share the computational power.



Computing paradigm shift

Electric Power Grid Analogy

Electric Power Grid

Users gets access to electricity through wall sockets with no care or consideration from where or how the electricity is actually generated.

The **power grid** links together power plants of many different kinds.

Grid Computing

- Users get access to computing resources as needed with little or no knowledge of where those resources are located or what the underlying technologies, operating system, hardware and so on are.
- The **Grid Links** together computing resources and provides mechanism to access those resources

Why it called as Grid?

Plugging into a computer grid will be as simple as plugging into **electric grid**.
Alike electric grid, users will **plug and use as much as computing power**.
User **need not to know** where the power is coming from.

When to use Grid Computing?

- **Share more than Information**: Data, Computing Power, multi-institutional needs.
- Efficient use of **resources** at **many institutes**. People in many institutes working on a **common problem**.
- Join local communities.

Need of Grid Computing?

- Today’s **research is more focused on computations**, data analysis and visualization.
- **Computer Simulation** is more **cost effective** than **actual experiments**.
- **Scientific problems** are getting more **complex, requiring more precise solutions in lesser time**.
- Exploiting underutilized resources.

Evolution of Cloud Computing

Several developments made possible the diffusion of computing grids:

Clusters became quite **common resources**.

Clusters were often **underutilized.**

New problems were requiring computational power that went beyond the capability of single clusters.

Improvement in internet technology, long-distance, high-bandwidth connectivity

Key	Cluster Computing	Grid Computing
Computer Type	Nodes or computers have to be of the same type.	Nodes or computers can be of the same or different types.
Task	Computers of Cluster Computing are dedicated to a single task.	Computers of Grid Computing can leverage unused computing resources to do other tasks.
Location	Computers of Cluster Computing are co-located.	Computers of Grid Computing can be present at different locations.
Topology	Centralized network topology.	De-centralized network topology.
Autonomy	In Cluster computing network, the whole system works as a unit.	In Grid computing network, each node is independent.
Task Scheduling	A centralized server controls the scheduling of tasks in cluster computing.	In Grid Computing, multiple servers can exist.
Resource Manager	Cluster Computing network has a dedicated centralized resource manager.	In Grid Computing, each node is independently managing each own resources.

Utility Computing

Def

- "*The packaging of computing resources (Computation, storage etc.) as a metered service similar to a traditional public utility*".

Observation

- Not a new concept
  - "If computers of the kind I have advocated become the computers of future, then computing may someday be organized as a public utility just as the telephone system is a public utility… The computer utility could become the basis of a new and important industry" – *John McCarthy, MIT Centennial in 1961*

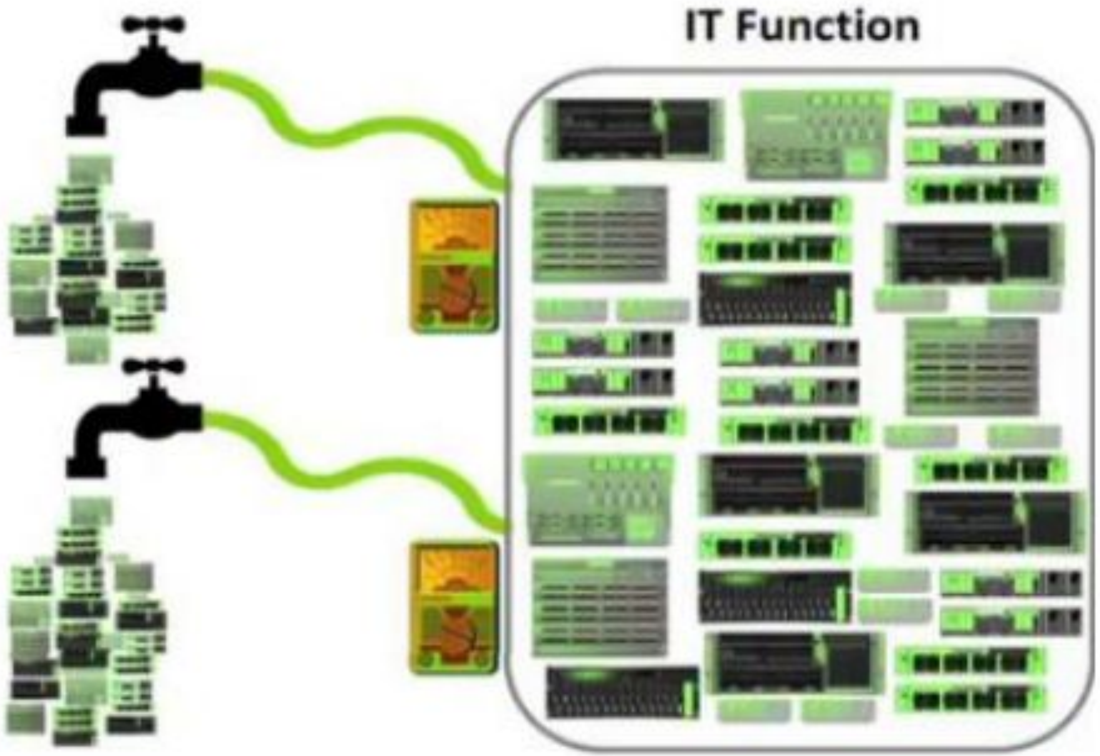
Utility Computing

- **Utility Computing** is a conceptual model practically implemented by cloud computing.
- Provides on-demand computing resources and infrastructure management.
- Charges users based on specific usage instead of a flat rate.
- Reduces or eliminates initial investment costs by allowing resource rental.
- The term "utility" is analogous to services like electricity, which charge based on consumption.
- Known for its **pay-per-use** approach, offering flexibility in meeting fluctuating customer needs.

Utility Computing

Utility Computing
has included some form of **Virtualization**;

The **amount of storage** or **computing** power **available** is considerably **larger** than that of a single time-sharing computer.



Source: Alan McSweeney, "The Economics of Utility Computing"

Utility Computing

- **Pay-per-use Pricing** model
- Solves **Resource Utilization** Problem
- **Outsourcing**
- **Web-based Services**
- **Automation**
- Data Center **Virtualization**

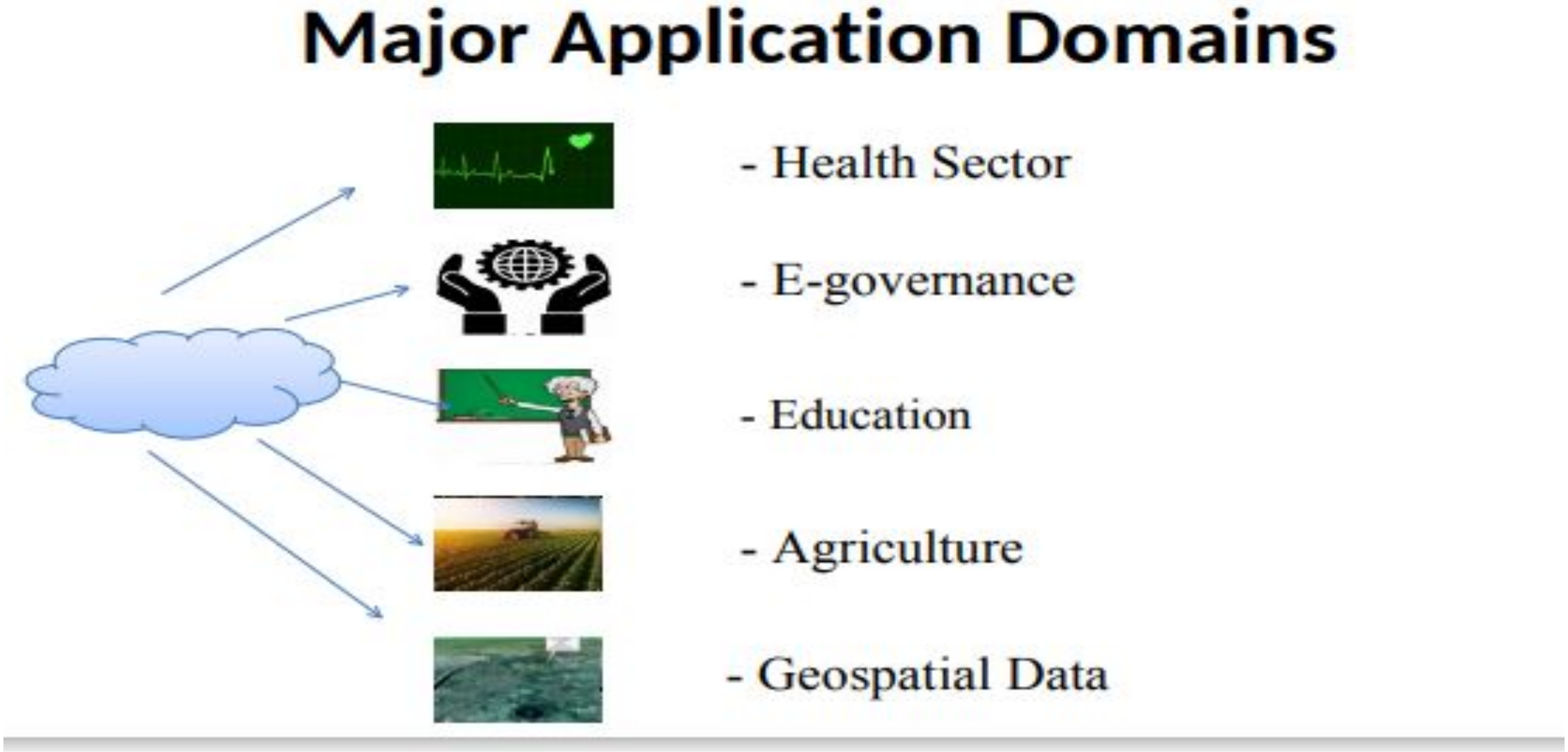
Utility Computing Payment Models

- Same range of charging models as other utility providers: gas, electricity, water, television broadcasting
  - **Flat Rate**: Fixed cost for unlimited usage within a defined scope.
  - **Pay per Use**: Charges based on actual resource consumption.
  - **Subscription**: Recurring fee for continuous access to services.
  - **Metered**: Billing tied to measurable usage metrics (e.g., GB/hour).
  - **Tiered**: Predefined packages with escalating features/costs.
- Different pricing models for different **customers** based on **factors** such as **scale, commitment, and payment frequency**.
- The **pricing model** is simply an expression by the provider of the **cost of provision of the resources** and a **profit margin**.

Autonomic Computing

- Autonomic Computing is a type of visionary computing that has been started by IBM. This is made to make adaptive decisions that use high-level policies. It has a feature of constant up-gradation using optimization and adaptation.
- IBM has defined the four areas of automatic computing:
  - Self-Configuration.
  - Self-Healing (error correction).
  - Self-Optimization (automatic resource control for optimal functioning).
  - Self-Protection (identification and protection from attacks in a proactive manner).

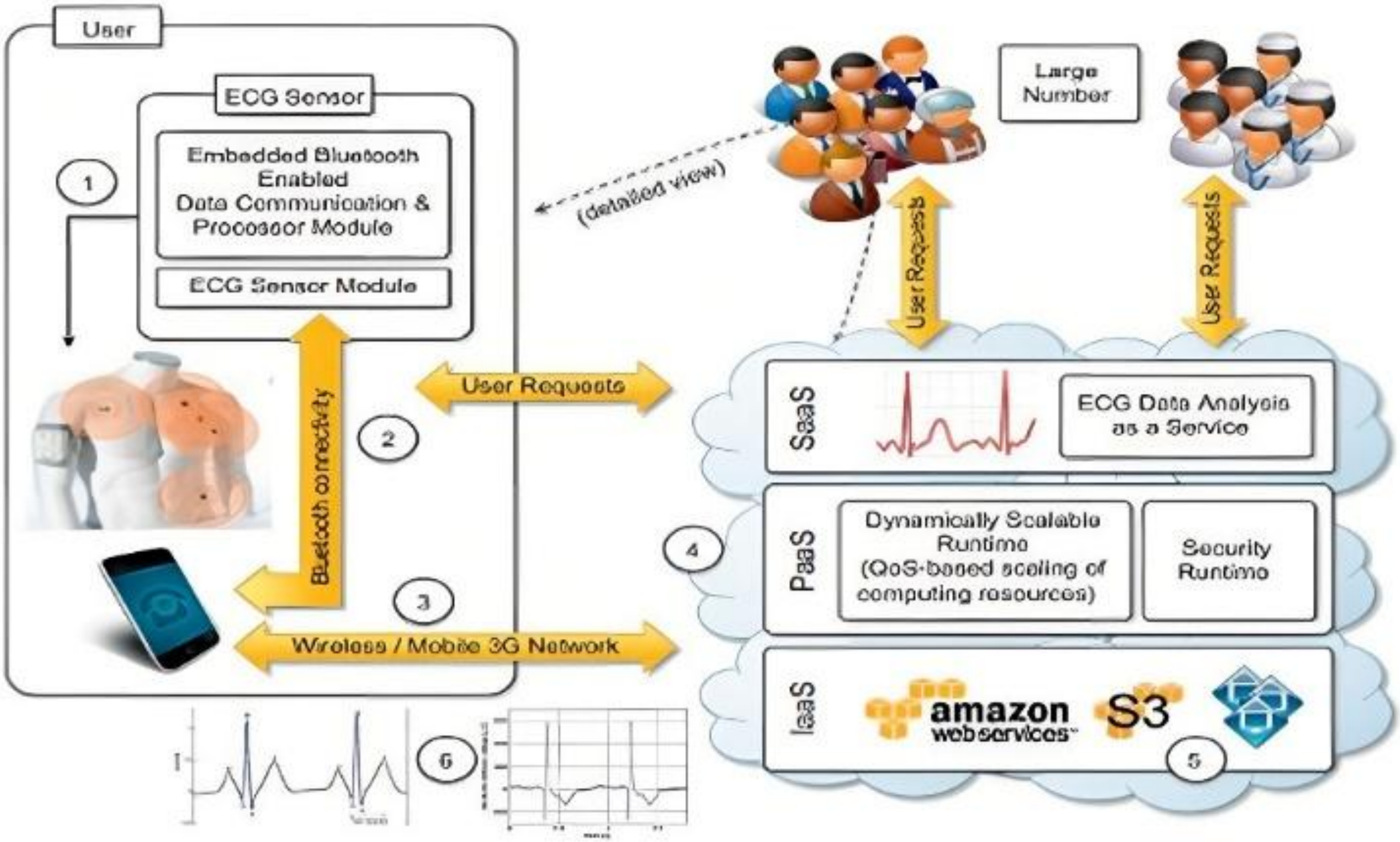
Cloud Computing Applications



Health Sector

- **ECG data analysis and monitoring**
  - ECG waveform is used to identify arrhythmias and is the most common way to detect heart disease.
  - **Cloud computing technologies Supports**
    - ✓ Remote monitoring of a patient's heartbeat.
    - ✓ Data analysis in minimal time.
    - ✓ notification of first-aid personnel and doctors

Health Sector



Health Sector

- Record medical history as well as current data, including health events, test results, vital stats, medications.

- Communicate this data to health care providers.

- Online medical facilities.

- **Example**: Windows Azure as service provider, National Health Portal India, etc.

E-Governance

- Cloud computing permits to uniformly cover the whole country with e-government solutions
- Service-oriented architecture facilitates provision of compound services.
- High order of customer order processing, where a customer may be a citizen or an enterprise

E-Governance Types

- Government to Government (G2G): Facilitates information exchange between different levels or branches of government.
  - e.g.: Crime and Criminal Tracking Network & Systems (CCTNS) – Links police stations across states and central agencies.

- Government to Citizen (G2C): Provides a platform for citizens to interact with the government and access public services.
  - e.g.: DigiLocker – Digital document storage for citizens linked to Aadhaar.

E-Governance Types

- Government to Business (G2B): Enhances interaction between businesses and the government for efficient access to business-related services.
  - E.g.: GeM (Government e-Marketplace) – Online procurement platform for government purchases.
- Government to Employees (G2E): Enables efficient communication between the government and its employees.
- E.g.:
  - SPARSH (System for Pension Administration – Raksha) – Pension management for defence personnel.
  - HRMS (Human Resource Management System) Indian Railways – Leave, service record, and salary management for railway employees.

Education

Cloud-based storage

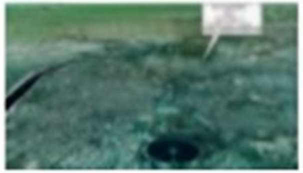
Virtual Hands-on Laboratories

Platform Services

Software Services

 Agriculture

- Soil Information
- Farmers' Data
- Expert Consultation
- E-commerce
- Weather Information



GeoScience Data

- **Geoscience** applications collect, produce, and analyze massive amounts of imaginary data.
- Process such data is both I/O and compute-intensive task.
- **Cloud computing** is an attractive option for executing these demanding tasks and extracting meaningful information to support decision makers.

Economics and Benefits of Cloud Computing

- **Economics of Cloud Computing:**
  - Based on the Pay-as-you-go model, where users pay only for the services they use.
- **Cost Benefits:**
  - Reduces indirect costs such as software licenses and support.
  - Users access software on a subscription basis, with the cloud provider maintaining ownership of the software.
- **Advantages for Developers:**
  - Pay-as-you-go model
  - Scalability
  - Simplicity
- **Cost Reductions:**
  - Reduces capital costs for infrastructure
  - Eliminates maintenance and administrative costs